14E022 Macroeconomics I

Practice Exam 1 — Answer Key

BSE — Academic Year 2025–26

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1 Question 1 — Ramsey Model with a Capital Income Tax

1. (5 points)

Solution:

Step 1: Resource constraint in per-effective-worker terms.
Starting from $\dot{K} = Y - C - \delta K$ (aggregate feasibility), divide both sides by AL:

$$\frac{\dot{K}}{AL} = \frac{Y}{AL} - \frac{C}{AL} - \delta\,\frac{K}{AL} = f(\tilde{k}) - \tilde{c} - \delta\tilde{k}.$$

Since $\tilde{k} = K/(AL)$, differentiating by the product rule:

$$\dot{\tilde{k}} = \frac{\dot{K}}{AL} - \frac{\dot{A}}{A}\tilde{k} - \frac{\dot{L}}{L}\tilde{k} = \frac{\dot{K}}{AL} - x\tilde{k} - n\tilde{k}.$$

Combining:

$$\boxed{\dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + x + \delta)\,\tilde{k} = \tilde{k}^{\alpha} - \tilde{c} - (n + x + \delta)\,\tilde{k}.}$$

Step 2: Transforming the household problem.
The per-capita budget constraint is $\dot{a} = (r(1-\tau) - n)a + w + T - c$. Defining $a = \tilde{k} \cdot A$ and $c = \tilde{c} \cdot A$ (so that $\dot{a} = \dot{\tilde{k}}A + \tilde{k}\dot{A} = (\dot{\tilde{k}} + x\tilde{k})A$), and dividing through by A:

$$\dot{\tilde{k}} + x\tilde{k} = \big(r(1-\tau) - n\big)\tilde{k} + \tilde{w} + \tilde{T} - \tilde{c},$$

$$\dot{\tilde{k}} = \big(r(1-\tau) - n - x\big)\tilde{k} + \tilde{w} + \tilde{T} - \tilde{c} = \big(f'(\tilde{k})(1-\tau) - \delta - n - x\big)\tilde{k} + \tilde{w} + \tilde{T} - \tilde{c},$$

where $\tilde{w} = w/A$ and $\tilde{T} = T/A$.

Step 3: Current-value Hamiltonian.
The household maximises $\int_0^\infty e^{-(\rho-n)t}\frac{c_t^{1-\sigma}-1}{1-\sigma}\,\mathrm{d}t$. Expressing utility in per-effective-worker terms ($c = \tilde{c}A$, $A = A_0 e^{xt}$):

$$c^{1-\sigma} = (\tilde{c}A)^{1-\sigma} = \tilde{c}^{1-\sigma}\,A_0^{1-\sigma}\,e^{(1-\sigma)xt}.$$

Absorbing the constant $A_0^{1-\sigma}$ and collecting exponentials, the effective discount rate in per-effective-worker form is $\rho' \equiv \rho - n - (1-\sigma)x$. The current-value Hamiltonian is:

$$\mathcal{H}^{CV} = \frac{\tilde{c}^{1-\sigma} - 1}{1-\sigma} + \lambda\big[\big(f'(\tilde{k})(1-\tau) - \delta - n - x\big)\tilde{k} + \tilde{w} + \tilde{T} - \tilde{c}\big].$$

State variable: $\tilde{k}$ (per-effective-worker capital).
Control variable: $\tilde{c}$ (per-effective-worker consumption).
Costate variable: λ (shadow value of an extra unit of $\tilde{k}$).
Note: Because the government rebates all tax revenue lump-sum ($T = \tau r k$), the *economy-wide resource constraint* is unaffected by τ. The tax only distorts the household's *perceived* return to saving.

2. (7 points)

Solution:

FOC w.r.t. $\tilde{c}$ ($\partial\mathcal{H}/\partial\tilde{c} = 0$):

$$\tilde{c}^{-\sigma} = \lambda. \tag{i}$$

Costate equation ($\dot{\lambda} = \rho'\lambda - \partial\mathcal{H}/\partial\tilde{k}$, where $\rho' = \rho - n - (1-\sigma)x$ is the effective discount

rate derived in part 1):

$$\frac{\partial \mathcal{H}}{\partial \tilde{k}} = \lambda\big[f'(\tilde{k})(1-\tau) - \delta - n - x\big].$$

Therefore:

$$\dot{\lambda} = \rho'\lambda - \lambda\big[f'(\tilde{k})(1-\tau) - \delta - n - x\big] = \lambda\big[\rho' - f'(\tilde{k})(1-\tau) + \delta + n + x\big]. \qquad \text{(ii)}$$

Eliminating the costate variable. Differentiating (i) with respect to time:

$$-\sigma\,\tilde{c}^{-\sigma-1}\,\dot{\tilde{c}} = \dot{\lambda} \quad \Longrightarrow \quad \frac{\dot{\lambda}}{\lambda} = -\sigma\,\frac{\dot{\tilde{c}}}{\tilde{c}}.$$

From (ii): $\dot{\lambda}/\lambda = \rho' - f'(\tilde{k})(1-\tau) + \delta + n + x$. Equating:

$$-\sigma\,\frac{\dot{\tilde{c}}}{\tilde{c}} = \rho' - f'(\tilde{k})(1-\tau) + \delta + n + x.$$

Substituting $\rho' = \rho - n - (1-\sigma)x$:

$$\begin{aligned}
-\sigma\,\frac{\dot{\tilde{c}}}{\tilde{c}} &= \big[\rho - n - (1-\sigma)x\big] - f'(\tilde{k})(1-\tau) + \delta + n + x \\
&= \rho - n - x + \sigma x - f'(\tilde{k})(1-\tau) + \delta + n + x \\
&= \rho + \sigma x + \delta - f'(\tilde{k})(1-\tau).
\end{aligned}$$

Dividing both sides by $-\sigma$:

$$\boxed{\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{1}{\sigma}\big[f'(\tilde{k})(1-\tau) - \delta - \rho - \sigma x\big].}$$

Effect of τ: The tax reduces the perceived net return from $f'(\tilde{k}) - \delta$ to $f'(\tilde{k})(1-\tau) - \delta$. Consumption growth is lower for any given $\tilde{k}$ compared to the no-tax case.
TVC: $\lim_{t\to\infty} \lambda(t)\,\tilde{k}(t)\,e^{-\rho' t} = 0$.

3. (7 points)

Solution:

(a) Modified Golden Rule: Setting $\dot{\tilde{c}}/\tilde{c} = 0$:

$$f'(\tilde{k}^*)(1-\tau) = \delta + \rho + \sigma x \quad \Longrightarrow \quad f'(\tilde{k}^*) = \frac{\delta + \rho + \sigma x}{1-\tau}.$$

With Cobb-Douglas $f(\tilde{k}) = \tilde{k}^\alpha$, $f'(\tilde{k}) = \alpha\tilde{k}^{\alpha-1}$:

$$\alpha(\tilde{k}^*)^{\alpha-1} = \frac{\delta + \rho + \sigma x}{1-\tau} \quad \Longrightarrow \quad \boxed{\tilde{k}^*(\tau) = \left[\frac{\alpha(1-\tau)}{\delta + \rho + \sigma x}\right]^{1/(1-\alpha)}.}$$

(b) Without tax ($\tau = 0$): $\tilde{k}^*(0) = \big[\alpha/(\delta + \rho + \sigma x)\big]^{1/(1-\alpha)}$.
Since $(1-\tau) < 1$ for $\tau > 0$:

$$\tilde{k}^*(\tau) = (1-\tau)^{1/(1-\alpha)}\,\tilde{k}^*(0) < \tilde{k}^*(0). \qquad \checkmark$$

(c) Intuition: Although the tax revenue is returned lump-sum, the household's *marginal* return to saving an extra unit of capital is reduced by the factor $(1-\tau)$. Lump-sum transfers are *inframarginal*: they add to wealth but do not change the return to saving at the margin. The household therefore saves less, and the economy converges to a lower capital stock.

4. (6 points)

Solution:

(a) Phase diagram:

- $\dot{\tilde{c}} = 0$ locus: vertical line at $\tilde{k}^*(\tau)$, which shifts *left* relative to $\tilde{k}^*(0)$ (since $\tilde{k}^*(\tau) < \tilde{k}^*(0)$).
- $\dot{\tilde{k}} = 0$ locus: $\tilde{c} = \tilde{k}^\alpha - (n + x + \delta)\tilde{k}$ (hump-shaped). This locus does *not* shift (the aggregate resource constraint is unaffected by τ; the tax only redistributes within the household).
- The new saddle path goes through the new steady state $(\tilde{k}^*(\tau), \tilde{c}^*(\tau))$, which lies on the $\dot{\tilde{k}} = 0$ curve to the left of the old steady state. Since the $\dot{\tilde{k}} = 0$ curve is hump-shaped and $\tilde{k}^*(\tau) < \tilde{k}^*(0) < \tilde{k}^*_{GR}$, the new $\tilde{c}^*$ is lower than the old $\tilde{c}^*$.

(b) Impact and dynamics:

- $\tilde{k}$ does **not** jump at $t = 0$ — capital is a state (predetermined) variable.
- $\tilde{c}$ **jumps up** at $t = 0$ to land on the new saddle path. The household now perceives a lower return to saving, so it immediately raises consumption.
- Transition: with $\tilde{c}$ above the $\dot{\tilde{k}} = 0$ locus, $\dot{\tilde{k}} < 0$ (capital decumulates). Meanwhile $\dot{\tilde{c}} < 0$ (we are to the right of the new $\dot{\tilde{c}} = 0$ locus). Both $\tilde{k}$ and $\tilde{c}$ gradually fall along the saddle path until the new steady state $(\tilde{k}^*(\tau), \tilde{c}^*(\tau))$ is reached.

2 Question 2 — AK Model and Learning-by-Doing Externalities

1. (5 points)

Solution:

(a) Each firm: $Y_j = \bar{A}\, K_j^{\alpha} L_j^{1-\alpha}$ with $\bar{A} = A_0 K^{\eta}$. Integrating over $j \in [0,1]$, using symmetry ($K_j = K$, $L_j = \bar{L}$):

$$Y = A_0\, K^{\eta+\alpha}\, \bar{L}^{1-\alpha}.$$

(b) With $\eta = 1-\alpha$ and $\bar{L} = 1$: $Y = A_0\, K^{(1-\alpha)+\alpha} \cdot 1 = A_0\, K \equiv AK$ where $A = A_0$.
Economic meaning: The externality parameter η exactly offsets diminishing returns to capital $(1-\alpha)$. Learning-by-doing generates enough knowledge spillovers to keep the aggregate marginal product of capital constant, removing the tendency for growth to slow down as capital accumulates.

2. (6 points)

Solution:

(a) With $\dot{K} = sY - \delta K = sAK - \delta K$:

$$\boxed{\gamma_K = \frac{\dot{K}}{K} = sA - \delta = \gamma_Y.}$$

Since $Y = AK$, $\gamma_Y = \gamma_K = sA - \delta$.
(b) Unlike Solow (where s affects only the *level* of y^*), here s directly enters the *growth rate* formula. A higher s raises γ_Y permanently. Intuition: with constant returns to capital ($f'' = 0$), more saving never runs into diminishing returns, so it sustains a permanently higher growth rate.
(c) Output per worker grows if $\gamma_Y - n > 0$, i.e. $sA - \delta > n$, or equivalently $sA > \delta + n$. (With $\bar{L}$ normalised to 1 and $n = 0$, the condition is simply $sA > \delta$.)

3. (7 points)

Solution:

(a) Setting up the household's Hamiltonian correctly.
A key subtlety: the question gives the equilibrium resource constraint $\dot{k} = (A-\delta)k - c$, but this combines *both* capital income *and* labour income. To derive the correct market Euler, we must set up the household's Hamiltonian using the *household's budget constraint* (in which the wage is taken as exogenous):

$$\dot{a} = (r - \delta)\, a + w - c,$$

where the household takes r and w as time paths given by the market. In the competitive equilibrium, firms pay $r = \alpha \bar{A} K^{\alpha-1} = \alpha A$ (private MPK before depreciation) and $w = (1-\alpha)\bar{A}K^{\alpha} = (1-\alpha)Ak$ (wage per worker, with $\bar{L} = 1$). In equilibrium $a = k$, so:

$$\dot{k} = (\alpha A - \delta)k + (1-\alpha)Ak - c = (A-\delta)k - c. \quad \checkmark$$

The equilibrium budget constraint and the resource constraint coincide, but the correct Hamiltonian for the household uses the return on *assets* only (wage is exogenous):

$$\mathcal{H}^{CV} = \frac{c^{1-\sigma} - 1}{1-\sigma} + \lambda\big[(\alpha A - \delta)\, a + w - c\big].$$

FOC $\partial\mathcal{H}/\partial c = 0$: $c^{-\sigma} = \lambda$.
Costate ($\dot{\lambda} = \rho\lambda - \partial\mathcal{H}/\partial a$): $\dot{\lambda} = \rho\lambda - \lambda(\alpha A - \delta)$, so $\dot{\lambda}/\lambda = \rho - (\alpha A - \delta)$.
Differentiating $c^{-\sigma} = \lambda$: $\dot{\lambda}/\lambda = -\sigma\dot{c}/c$.
Combining:

$$-\sigma\,\frac{\dot{c}}{c} = \rho - (\alpha A - \delta) \quad\Longrightarrow\quad \boxed{\frac{\dot{c}}{c} = \frac{\alpha A - \delta - \rho}{\sigma} \equiv \gamma_c^{\text{mkt}}.}$$

(b) The household faces the *private* return $r = \alpha A$ because each competitive firm takes $\bar{A} = A_0 K^{1-\alpha}$ as given and sets $r = \partial Y_j/\partial K_j = \alpha\bar{A}K_j^{\alpha-1} = \alpha A$ in equilibrium. The household *does not* internalise that its own saving raises $\bar{A}$, so it only earns the private return $\alpha A < A$ (the social return), giving a lower market growth rate than the planner's.
(c) TVC: $\lim_{t\to\infty} \lambda(t)k(t)e^{-\rho t} = 0$ requires $\rho > (1-\sigma)\gamma_c^{\text{mkt}}$, ensuring bounded lifetime utility.

4. (7 points)

Solution:

(a) The planner internalises $\bar{A} = A_0 K^{1-\alpha}$ and sees social MPK $= A$ (since $Y = AK$ at the aggregate level).
Planner's Euler:

$$\gamma_c^{\text{plnr}} = \frac{A - \delta - \rho}{\sigma}.$$

Since $\alpha < 1$: $\gamma_c^{\text{plnr}} - \gamma_c^{\text{mkt}} = \frac{(1-\alpha)A}{\sigma} > 0$.
Externality: The market under-invests because firms do not internalise the positive knowledge spillover: each firm's capital accumulation raises aggregate $\bar{A}$ for all firms, but this benefit is external and uncompensated.
(b) To restore the planner's allocation, subsidise the return to capital by s^* so that the private perceived return equals the social return:

$$\alpha A(1 + s^*) = A \quad\Longrightarrow\quad \boxed{s^* = \frac{1-\alpha}{\alpha}.}$$

Interpretation: the subsidy rate equals the ratio of the external benefit $(1-\alpha)A$ to the private return αA. For $\alpha = 1/3$: $s^* = 2$, a 200% subsidy.
(c) Practical challenge: the government must know α precisely (or equivalently, the magnitude of the externality η) to set the subsidy correctly. If the externality is over-estimated, the subsidy induces over-accumulation. Additionally, financing the subsidy requires taxation, which may create its own distortions.

3 Question 3 — Quality Ladders and Schumpeterian Growth

1. (5 points)

Solution:

Final good firm maximises:

$$\max_{L_Y,\{x_s\}} L_Y^{1-\alpha} \int_0^N (q^{k(s)} x_s)^\alpha \, \mathrm{d}s - wL_Y - \int_0^N p_s \, x_s \, \mathrm{d}s.$$

FOC for x_s:

$$\alpha L_Y^{1-\alpha} (q^{k(s)})^\alpha \, x_s^{\alpha-1} = p_s \quad \Longrightarrow \quad \boxed{x_s = \left(\frac{\alpha L_Y^{1-\alpha} (q^{k(s)})^\alpha}{p_s} \right)^{1/(1-\alpha)}}.$$

Demand is isoelastic with elasticity $\varepsilon = 1/(1-\alpha)$.

2. (6 points)

Solution:

(a) The monopolist in sector s (with quality $q^{k(s)}$, MC $= 1$) chooses x_s to maximise profit:

$$\pi_s = (p_s - 1)\, x_s,$$

subject to the final-good firm's demand $x_s = \left(\alpha L_Y^{1-\alpha}(q^{k(s)})^\alpha / p_s\right)^{1/(1-\alpha)}$, or equivalently $p_s = \alpha L_Y^{1-\alpha}(q^{k(s)})^\alpha x_s^{\alpha-1}$.
Substituting into profit and differentiating with respect to x_s (treating demand as the inverse-demand curve):

$$\pi_s = p_s(x_s)\, x_s - x_s = \alpha L_Y^{1-\alpha}(q^{k(s)})^\alpha\, x_s^\alpha - x_s.$$

$$\frac{\partial \pi_s}{\partial x_s} = \alpha^2 L_Y^{1-\alpha}(q^{k(s)})^\alpha\, x_s^{\alpha-1} - 1 = \alpha\, p_s - 1 = 0 \quad \Longrightarrow \quad p_s^* = \frac{1}{\alpha}.$$

Equivalently, using the isoelastic demand elasticity $\varepsilon = 1/(1-\alpha)$, the optimal markup is $p^* = \varepsilon/(\varepsilon - 1) \cdot MC = (1/(1-\alpha))/(\alpha/(1-\alpha)) \cdot 1 = \boxed{1/\alpha}$.
(b) Limit pricing: The previous-generation producer has quality $q^{k(s)-1}$ and $MC = 1$. A consumer buying from the old firm gets value $\propto q^{k(s)-1}/p_{\text{old}}$. The new monopolist (quality $q^{k(s)}$) can charge up to $p = q$ before the old firm undercuts. Since $q \geq 1/\alpha$ and the unconstrained monopoly price is $1/\alpha$, we have $1/\alpha \leq q$, so the limit-pricing constraint is *not binding*.
(c) From the final good firm's equilibrium with symmetric firms ($x_s = x$):

$$p \cdot x = \frac{\alpha Y}{N} \quad \text{(total intermediate payments} = \alpha Y).$$

Profits:

$$\pi_s = (p-1)x = \left(\frac{1}{\alpha} - 1\right) x = \frac{1-\alpha}{\alpha}\, x.$$

Since $px = \alpha Y/N$ and $p = 1/\alpha$: $x = \alpha^2 Y/N$, so

$$\boxed{\pi_s = \frac{(1-\alpha)}{\alpha} \cdot \frac{\alpha^2 Y}{N} = \frac{\alpha(1-\alpha)\, Y}{N}.}$$

3. (7 points)

Solution:

(a) On the BGP, profits grow at rate g_Y (since $\pi \propto Y$). The value V_s also grows at rate g_Y. The no-arbitrage (Bellman) equation for V_s:

$$r\, V_s = \pi_s + \dot{V}_s - \iota\, V_s = \pi_s + g_Y V_s - \iota V_s.$$

Solving:

$$\boxed{V_s = \frac{\pi_s}{r + \iota - g_Y}.}$$

Interpretation: the "effective discount rate" is $r + \iota - g_Y$, augmented by ι (risk of displacement through creative destruction) and reduced by g_Y (capital gains from growing profits).
(b) Free entry into R&D: a firm spending h units of final output gets innovation probability μh. Zero profit requires:

$$\mu V_s = 1 \quad \Longrightarrow \quad V_s = \frac{1}{\mu}.$$

Combining: $\pi_s/(r+\iota-g_Y) = 1/\mu$, so $r = \mu\pi_s - \iota + g_Y$.
(c) The household Euler equation is $\dot{c}/c = r - \rho$ (with $\sigma = 1$ for simplicity, or more precisely $r - \rho = g_Y$ on the BGP where $\dot{c}/c = g_Y$). On the BGP by definition $\dot{c}/c = g_Y$, so:

$$g_Y = r - \rho \quad \Longrightarrow \quad r = \rho + g_Y.$$

From part (b), the free-entry condition gives $r = \mu\pi_s - \iota + g_Y$. Substituting $r = \rho + g_Y$:

$$\rho + g_Y = \mu\pi_s - \iota + g_Y.$$

The g_Y terms cancel, leaving:

$$\boxed{\iota = \mu\pi_s - \rho.}$$

The BGP growth rate of output is $g_Y = \iota \ln q$ (each innovation raises quality by factor q in one sector, and innovations arrive at rate ι per sector):

$$\boxed{g_Y = (\mu\pi - \rho)\ln q.}$$

4. (7 points)

Solution:

(a) Business-stealing (creative destruction) externality: When a new innovator succeeds, it destroys the incumbent monopolist's profit stream. The entrant does not account for this loss imposed on the incumbent. ⇒ **Over-innovation** relative to social optimum: the private incentive to innovate is inflated because the entrant captures the *entire* market value, ignoring the rents it destroys.
(b) Appropriability (consumer-surplus) externality: The innovator cannot appropriate all consumer surplus from the quality improvement. Consumers benefit from higher quality beyond what the monopolist extracts through pricing. ⇒ **Under-innovation**: the private value of innovation is less than the social value because the innovator's profits understate the total welfare gain.
(c) In the Aghion–Howitt model, *either* externality can dominate, depending on parameters (q, α, σ). When q is large (drastic innovations), business stealing is severe and the market may over-innovate. When innovations are incremental, the appropriability effect may dominate, leading to under-innovation. In general, a single instrument (e.g., an R&D subsidy or tax) may not achieve the first-best, because correcting one externality may worsen the other. Optimal policy may require *two* instruments: an R&D subsidy/tax to correct the innovation rate, and a production subsidy to correct the static under-use of intermediates.

4 Short Questions

1. (9 points) — Convergence

Solution:

(a) Absolute convergence: poorer countries grow faster than richer countries regardless of their structural parameters. **Conditional convergence**: each country converges to its own steady state; poorer countries grow faster *conditional on* having the same structural parameters (s, n, δ, etc.).
The basic Solow model predicts **conditional convergence** (not absolute), because steady states differ across countries. "Conditioning" means comparing countries with similar savings rates, population growth, and technology.
(b) Yes. US states share similar institutions, savings behaviour, and technology access, so they approximately share the same steady state $\Rightarrow$ we observe conditional (and approximately absolute) convergence among them. In the cross-country sample, steady states differ widely $\Rightarrow$ the unconditional correlation is weak. This is exactly what the Solow model predicts.
(c) The speed of convergence around the steady state:

$$\beta = (1-\alpha)(n+g+\delta).$$

β governs the rate at which $\ln \tilde{k}$ closes the gap to $\ln \tilde{k}^*$. A higher α (capital share) slows convergence (diminishing returns kick in more slowly); higher n, g, or δ speed it up (larger effective depreciation of per-effective-worker capital).

2. (8 points) — Scale Effects and the Jones Fix

Solution:

(a) In Romer (1990), the R&D equation is $\dot{M} = \lambda M L_R$. More R&D workers L_R generate more varieties, raising the growth rate. Since L_R is proportional to population L, larger economies grow permanently faster. Empirically, this is problematic: the number of scientists has grown enormously since 1950, yet growth rates in developed countries have been roughly constant.
(b) Jones replaces $\dot{M} = \lambda M L_R$ with $\dot{M} = \lambda M^{\phi} L_R$ where $\phi < 1$. On the BGP, M grows at rate g_M and L_R grows at rate n (population growth):

$$g_M = \frac{\dot{M}}{M} = \lambda M^{\phi-1} L_R.$$

For g_M to be constant, $(\phi - 1)g_M + n = 0$, so

$$\boxed{g_M = \frac{n}{1-\phi}.}$$

The growth rate depends on population *growth* n, not population *level* L. Scale effect disappears. If $n = 0$, long-run growth is zero — sustained growth requires positive population growth.
(c) Key criticism: growth becomes "semi-endogenous" — it depends on the exogenous demographic parameter n. Policy (R&D subsidies, patent protection) cannot affect the long-run growth rate, only the level. This contradicts the motivation for endogenous growth theory and limits its policy relevance.

3. (8 points) — Non-Homotheticity and Structural Change

Solution:

(a) Preferences are **non-homothetic** when the ratio of goods demanded changes as income increases, holding prices fixed. Formally, the income elasticity of demand $\varepsilon_I \neq 1$ for at least one good. As economies grow richer, households shift spending toward goods with $\varepsilon_I > 1$ (luxuries, e.g. services) and away from goods with $\varepsilon_I < 1$ (necessities, e.g. food). This demand-side reallocation drives structural change.
(b) With Cobb-Douglas aggregation, expenditure shares are constant by construction (regardless of income). To generate a rising service share, one must add either: (i) **non-homothetic preferences** (e.g. Stone-Geary with subsistence requirements), or (ii) **differential productivity growth** across sectors with elasticity of substitution < 1 (supply-side structural change). Concrete example: Stone-Geary utility $U = (c_M - \bar{c}_M)^{\beta} c_S^{1-\beta}$ with subsistence $\bar{c}_M > 0$; as income rises, the share spent on M above subsistence falls.
(c) Baumol's "cost disease": if manufacturing has faster productivity growth than services, and goods are complementary (low substitution elasticity), the relative price of services rises. At constant nominal expenditure shares, the *employment* share in services rises because services require ever more labour per unit of output. For aggregate growth: the economy's overall productivity growth is a weighted average, with increasing weight on the stagnant sector $\Rightarrow$ aggregate growth slows over time ("Baumol's disease").

End of Answer Key — Practice Exam 1.